

LOCAL CACCIOPPOLI ESTIMATES AND THE ROUGH-CORE ALTERNATIVE FOR TYPE II NAVIER-STOKES RESCALINGS

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ABSTRACT. We prove a local regularity criterion for Type II Navier–Stokes rescalings in an absolutely continuous scale-translation gauge. The assumptions are physical suitability, pressure reconstruction modulo functions of time, a bounded critical L^3 norm, bounded modulation coefficients, and the local renormalized Caccioppoli inequality on compact cylinders. Under these assumptions the renormalized velocity has uniform local windowed H^1 bounds on every bounded renormalized ball. Consequently an L^3 -tight Type II renormalization satisfying these hypotheses cannot exhibit persistent local high-frequency activity in a bounded renormalized core. We also record the corresponding formulation with modulation controlled in L^1 on selected enlarged windows.

1. INTRODUCTION

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

near a putative Type II singular time. Suppose that, along a terminal concentration scale and center, the solution has been written in renormalized variables (V, P) , with modulation coefficients a, b , on a half-line $[\tau_0, \infty)$. The local question addressed here is whether a bounded critical renormalized core can retain persistent high-frequency activity while remaining L^3 -controlled. The answer is negative under the analytic hypotheses stated below.

The result is the following local estimate. Suppose the renormalized solution satisfies

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0,$$

with pressure determined by

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

modulo functions of τ . If V is bounded in $L^\infty_\tau L^3_y$, the modulation coefficients a, b are bounded, and the renormalized Caccioppoli inequality is valid on compact cylinders, then

$$\sup_{T \geq \tau_0+1} \int_T^{T+1} \|V(\tau)\|_{H^1(B_R)}^2 d\tau < \infty \quad \text{for every } R > 0.$$

This excludes the rough-core alternative, understood here as failure of these local windowed H^1 bounds inside a bounded renormalized core.

Theorem 1.1 (Main local regularity criterion). *Let (V, P, a, b) solve the renormalized Navier–Stokes equation above on $[\tau_0, \infty) \times \mathbb{R}^3$, in the sense of the local energy inequality on compact cylinders. Assume that*

$$\sup_{\tau \geq \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty, \quad \sup_{\tau \geq \tau_0} (|a(\tau)| + |b(\tau)|) < \infty,$$

and that

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

modulo functions of τ . Then, for every $R > 0$,

$$\sup_{T \geq \tau_0 + 1} \int_T^{T+1} \|V(\tau)\|_{H^1(B_R)}^2 d\tau < \infty.$$

Consequently the rough-core alternative, defined by failure of this estimate on some fixed ball B_R , cannot occur under these hypotheses.

Physical suitability is understood in the sense of Caffarelli–Kohn–Nirenberg [CKN82]. The pressure estimates use Calderon–Zygmund/Riesz-transform theory [CZ52, Ste70]. No global energy bound at infinity is used; the Caccioppoli step is local on compact renormalized cylinders.

2. SUITABLE SOLUTIONS AND RENORMALIZED LOCAL ENERGY

Definition 2.1 (Physical suitability). A pair (u, p) is physically suitable on a spacetime cylinder if

$$u \in L_t^\infty L_{\text{loc}}^2 \cap L_t^2 H_{\text{loc}}^1, \quad p \in L_{\text{loc}}^{3/2}, \quad \nabla \cdot u = 0,$$

the Navier–Stokes equations hold distributionally, and for every nonnegative $\psi \in C_c^\infty$ one has the local energy inequality

$$\begin{aligned} & \int \frac{|u(t_2)|^2}{2} \psi(t_2) + \nu \int_{t_1}^{t_2} \int |\nabla u|^2 \psi \\ & \leq \int \frac{|u(t_1)|^2}{2} \psi(t_1) + \int_{t_1}^{t_2} \int \frac{|u|^2}{2} (\partial_t \psi + \nu \Delta \psi) + \int_{t_1}^{t_2} \int \left(\frac{|u|^2}{2} + p \right) u \cdot \nabla \psi. \end{aligned}$$

Definition 2.2 (Absolutely continuous renormalized chart). An absolutely continuous renormalized chart on a compact renormalized time interval has the form

$$u(x, t) = \Lambda(s)^{-1} V(Y, s), \quad p(x, t) = \Lambda(s)^{-2} P(Y, s) + c(t),$$

where

$$Y = \frac{x - X(s)}{\Lambda(s)}, \quad dt = \Lambda(s)^2 ds,$$

with $\Lambda > 0$ and $X, \Lambda \in W_{\text{loc}}^{1,1}$. The final modulation coefficients are denoted

$$a(s) = \partial_s \log \Lambda(s), \quad b(s) = \Lambda(s)^{-1} X_s(s),$$

where derivatives are interpreted a.e.

Lemma 2.3 (Pullback of the local energy inequality). *Let (u, p) be physically suitable on a compact physical cylinder and let (V, P) be obtained from an absolutely continuous renormalized chart. Then (V, P) satisfies, on compact renormalized cylinders, the local energy inequality*

$$\begin{aligned} & \int \frac{|V(s_2)|^2}{2} \phi(s_2) + \nu \int_{s_1}^{s_2} \int |\nabla V|^2 \phi \\ & \leq \int \frac{|V(s_1)|^2}{2} \phi(s_1) + \int_{s_1}^{s_2} \int \frac{|V|^2}{2} (\partial_s \phi + \nu \Delta \phi) \\ & \quad + \int_{s_1}^{s_2} \int \left(\frac{|V|^2}{2} + P \right) V \cdot \nabla \phi \\ & \quad + \int_{s_1}^{s_2} \int a(s) \frac{|V|^2}{2} (-\phi - y \cdot \nabla \phi) - \int_{s_1}^{s_2} \int \frac{|V|^2}{2} b(s) \cdot \nabla \phi \end{aligned}$$

for every nonnegative $\phi \in C_c^\infty(\mathbb{R}^3 \times (s_1, s_2))$.

Proof. First suppose X and Λ are C^1 . Apply the physical local energy inequality with

$$\psi(x, t) = \phi\left(\frac{x - X(s)}{\Lambda(s)}, s\right), \quad t = t(s).$$

The support remains compact because Λ is bounded above and below on the selected compact interval. Changing variables $x = X(s) + \Lambda(s)y$ and $dt = \Lambda^2 ds$, and using $u = \Lambda^{-1}V$, $p = \Lambda^{-2}P + c(t)$, gives the displayed inequality. The pressure constant drops out since

$$\int c(t)V \cdot \nabla \phi = - \int c(t)\phi \nabla \cdot V = 0.$$

The terms involving Λ_s and X_s are exactly the scale and translation terms with coefficients a and b .

For $X, \Lambda \in W_{\text{loc}}^{1,1}$, mollify the pulled-back test function. The suitability classes give the integrability needed for every term in the local energy inequality, and the mollified tests may be chosen with support in a slightly larger compact cylinder. Passing to the limit gives the same renormalized inequality. Only the locally integrable functions X_s and Λ_s appear in the additional terms. $\square$

3. THE LOCAL CACCIOPPOLI ESTIMATE

Lemma 3.1 (Renormalized Caccioppoli inequality). *Assume (V, P, a, b) satisfies the renormalized local energy inequality on the compact cylinder*

$$Q_{2R,I} = (T-1, T+2) \times B_{2R}.$$

Assume also that a, b are bounded on this interval and that

$$V \in L^\infty(I; L^3(B_{2R})), \quad P \in L^{3/2}(Q_{2R,I}).$$

Then

$$\begin{aligned} \nu \int_T^{T+1} \int_{B_R} |\nabla V|^2 &\leq C_R \iint_{Q_{2R,I}} |V|^2 + C_R \iint_{Q_{2R,I}} |V|^3 \\ &\quad + C_R \left| \iint_{Q_{2R,I}} P V \cdot \nabla \zeta \right| + C_R \|a, b\|_{L^\infty(I)} \iint_{Q_{2R,I}} |V|^2, \end{aligned}$$

where ζ is a standard spatial cutoff equal to one on B_R , supported in B_{2R} , and satisfying $|\nabla \zeta| + |\Delta \zeta| \leq C_R$.

Proof. Insert a cutoff $\phi(y, \tau) = \eta(\tau)\zeta(y)^2$ in the renormalized local energy inequality, with η supported in $(T-1, T+2)$ and equal to one on $[T, T+1]$. The viscous term controls $\nu \int_T^{T+1} \int_{B_R} |\nabla V|^2$. The terms involving $\partial_\tau \phi$, $\Delta \phi$, and $V \cdot \nabla \phi$ give the local L^2 and L^3 contributions. The pressure term remains in the displayed form because pressure is defined modulo functions of time. The scale and translation terms are controlled by the local L^2 -mass multiplied by the bounded modulation norm. This is the stated estimate. $\square$

4. PRESSURE CONTROL FROM THE CRITICAL NORM

Lemma 4.1 (Local pressure estimate modulo constants). *Assume*

$$-\Delta P(\tau) = \partial_i \partial_j (V_i(\tau) V_j(\tau)) \quad \text{in } \mathbb{R}^3$$

for a.e. τ , and

$$M_3 := \sup_{\tau \geq \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Then for every $R > 0$ there are measurable constants $c_R(\tau)$ such that

$$\sup_{\tau \geq \tau_0} \|P(\tau) - c_R(\tau)\|_{L^{3/2}(B_R)} \leq C_R M_3^2.$$

Proof. Write the pressure as the local singular-integral part plus the harmonic tail. For the local part, Calderon–Zygmund estimates give

$$\|\mathcal{R}_i \mathcal{R}_j (V_i V_j)\|_{L^{3/2}(B_{4R})} \leq C \|V\|_{L^3(B_{4R})}^2.$$

For the far-field part, subtract its value at the center of the ball. The kernel difference is bounded by $C_R |z|^{-4}$ for $|z| \geq 2R$, giving a harmonic contribution controlled by the usual pressure tail

$$\sup_{x \in B_R} \int_{|z| > 2R} \frac{|V(z, \tau)|^2}{|x - z|^5} dz \leq C_R M_3^2.$$

The subtracted value is the constant $c_R(\tau)$. Combining the two bounds gives the estimate. The constants may be chosen measurably by fixing this pressure decomposition. $\square$

5. UNIFORM WINDOWED H^1 CONTROL

Theorem 5.1 (Windowed local gradient estimate). *Let (V, P, a, b) be a renormalized Navier–Stokes solution on $[\tau_0, \infty)$. Assume:*

- (i) *the renormalized local energy inequality holds on compact cylinders;*
- (ii)

$$M_3 := \sup_{\tau \geq \tau_0} \|V(\tau)\|_{L^3(\mathbb{R}^3)} < \infty;$$

(iii)

$$M_{ab} := \sup_{\tau \geq \tau_0} (|a(\tau)| + |b(\tau)|) < \infty;$$

- (iv) *the pressure satisfies $-\Delta P = \partial_i \partial_j (V_i V_j)$ modulo functions of time.*

Then for every $R > 0$ there is a constant $C_R = C(R, \nu, M_3, M_{ab})$ such that

$$\sup_{T \geq \tau_0 + 1} \int_T^{T+1} \int_{B_R} |\nabla V(y, \tau)|^2 dy d\tau \leq C_R.$$

Proof. Fix $R > 0$ and $T \geq \tau_0 + 1$. Apply the Caccioppoli estimate on $(T - 1, T + 2) \times B_{2R}$. The local L^2 -term is bounded by Holder's inequality:

$$\int_{B_{2R}} |V(\tau)|^2 dy \leq |B_{2R}|^{1/3} \|V(\tau)\|_{L^3(\mathbb{R}^3)}^2 \leq C_R M_3^2.$$

The outer time interval has length three, so the spacetime L^2 -term is bounded by $C_R M_3^2$. Similarly,

$$\iint_{(T-1, T+2) \times B_{2R}} |V|^3 \leq 3M_3^3.$$

It remains to estimate the pressure contribution. Since constants vanish in $\int PV \cdot \nabla \zeta$, replace P by $P - c_{2R}(\tau)$. Holder's inequality and the pressure estimate give, for a.e. τ ,

$$\begin{aligned} \int_{B_{2R}} |P - c_{2R}(\tau)| |V(\tau)| dy &\leq \|P(\tau) - c_{2R}(\tau)\|_{L^{3/2}(B_{2R})} \|V(\tau)\|_{L^3(B_{2R})} \\ &\leq C_R M_3^3. \end{aligned}$$

Integrating in τ over an interval of length three gives the same bound up to the constant C_R . The modulation term is controlled by $C_R M_{ab} M_3^2$. Substituting these estimates into the Caccioppoli inequality and dividing by ν proves the claim. $\square$

Corollary 5.2 (Uniform local windowed H^1 bound). *Under the hypotheses of the preceding theorem, for every $R > 0$,*

$$\sup_{T \geq \tau_0+1} \int_T^{T+1} \|V(\tau)\|_{H^1(B_R)}^2 d\tau < \infty.$$

Proof. The gradient part is the preceding theorem. The local L^2 -part follows from the same bounded-set estimate:

$$\int_T^{T+1} \|V(\tau)\|_{L^2(B_R)}^2 d\tau \leq C_R \int_T^{T+1} \|V(\tau)\|_{L^3(\mathbb{R}^3)}^2 d\tau \leq C_R M_3^2.$$

Adding the two bounds gives the result. $\square$

6. ROUGH-CORE EXCLUSION

Definition 6.1 (Rough-core behavior). A renormalized Type II solution satisfying the bounded critical norm and pressure reconstruction hypotheses has rough-core behavior if there exists $R > 0$ such that

$$\sup_{T \geq \tau_0+1} \int_T^{T+1} \|V(\tau)\|_{H^1(B_R)}^2 d\tau = \infty.$$

Theorem 6.2 (Exclusion of rough-core behavior). *Let (V, P, a, b) be a renormalized Type II solution satisfying the hypotheses of the windowed local gradient estimate. Then rough-core behavior is impossible.*

Proof. The corollary gives a finite windowed H^1 bound on every ball B_R . This is incompatible with the definition of rough-core behavior. $\square$

7. INTEGRATED MODULATION VARIANT

The pointwise boundedness of a and b can be weakened on selected windows when the argument only requires Caccioppoli estimates on those windows. We record the corresponding local formulation.

Assumption 7.1 (Integrated modulation control on enlarged windows). For every selected window $[T, T+1]$, assume that the enlarged window $[T-1, T+2]$ is contained in the renormalized time interval and that

$$\int_{T-1}^{T+2} (|a(\tau)| + |b(\tau)|) d\tau \leq M_{ab}^{(1)}.$$

Proposition 7.2 (Caccioppoli estimate with integrated modulation). *Assume the hypotheses of the windowed local gradient estimate, except replace pointwise boundedness of a, b by integrated modulation control on enlarged windows. Then, for every selected window $[T, T+1]$ and every $R > 0$,*

$$\int_T^{T+1} \int_{B_R} |\nabla V(y, \tau)|^2 dy d\tau \leq C_R,$$

where C_R depends on R, ν, M_3 , and $M_{ab}^{(1)}$, but not on T .

Proof. Use the same Caccioppoli cutoff as in the proof of the windowed local gradient estimate: it is supported in $(T-1, T+2) \times B_{2R}$ and equals one on $[T, T+1] \times B_R$. The proof is unchanged except for the modulation term. In the Caccioppoli inequality this term has the form

$$\int_{T-1}^{T+2} (|a(\tau)| + |b(\tau)|) \int_{B_{2R}} |V(y, \tau)|^2 dy d\tau.$$

The global L^3 -bound gives

$$\int_{B_{2R}} |V(y, \tau)|^2 dy \leq C_R M_3^2$$

for a.e. τ . The integrated modulation hypothesis therefore bounds the whole modulation contribution by $C_R M_3^2 M_{ab}^{(1)}$. The remaining terms are estimated exactly as before. $\square$

Remark 7.3. This integrated variant is not a substitute for pointwise bounded modulation when estimates are needed uniformly on all unit windows. It applies to selected windows for which the modulation coefficients are controlled in L^1 on the corresponding enlarged window.

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